

Pasindu Bandarigoda

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A researcher-practitioner with expertise in software engineering, AI, and data-driven healthcare systems. My goal is to pursue a PhD that advances personalized, privacy-preserving, and ethically aligned AI for health informatics and bioinformatics applications. I am particularly interested in developing culturally adaptive and safe conversational AI for clinical environments, leveraging my experience in machine learning, edge deployment, and intelligent system design.

Education:

- Master of Science, Software Engineering University of Westminster, Sep 2022 - Aug 2024 - Merit Pass
- Bachelor of Science (Hons), Information Technology Specializing in Information Technology, SLIIT, 2018 - Mar 2022

Relevant Coursework:

- Machine Learning (BSc MSc)
- Deep Learning (BSc)
- Data Science & Analytics (BSc MSc)
- Software Engineering (BSc MSc)
- Cloud Computing (BSc MSc)

Research Experience:

- **BrAIN Labs Inc.: Ongoing Research**
 - **Towards Tiny Transformers:** Investigating transformer model compression and architecture optimization to enable efficient NLP, vision, and forecasting inference on edge devices.
 - **Digital Phenotyping and AI for Mental Health Monitoring:** Exploring smartphone-based behavioral sensing and AI-driven mental health assessment, with emphasis on privacy-preserving data collection, adaptive participant modeling, and ethical deployment.
- **MSc Research: Enhancing MRI Image Segmentation Through Quantization and AI-Powered Algorithms for Clinical Efficiency**
 - Developed a quantized U-Net model for MRI brain tumor segmentation to enable real-time clinical analysis on mainstream hardware.
 - Reduced model size from 124 MB to 31 MB and improved inference speed from 1.2 s to 0.4 s per MRI volume while maintaining a Dice score of 0.89.
 - Demonstrated how model quantization can enhance deployment feasibility and energy efficiency in healthcare AI systems.
- **BSc Research: CCTV-Based Enhanced Public Security Management System for Sri Lanka**
 - Conducted applied research in collaboration with the Sri Lanka Police CCTV Division, utilizing authentic surveillance data under a signed NDA.
 - Designed and trained YOLO-based deep learning models for real-time detection of fires, accidents, and traffic violations.
 - Created a custom dataset exceeding 10,000 labeled images and implemented a cloud-based GCP architecture for live video analytics.
 - Achieved a 70% reduction in manual monitoring workload, enhancing the responsiveness of public security operations.

Publications:

"CCTV based Enhanced Public Security Management System for Sri Lanka" Published in IJCA – Foundation of Computer Science (FCS), NY, USA, Oct 2021

Research Interests:

- Health informatics and bioinformatics-inspired approaches for intelligent healthcare systems
- Privacy-preserving and decentralized machine learning, including federated learning
- Model quantization, optimization, and deployment of efficient AI in real-world applications
- Personalized AI solutions for clinical decision support and digital well-being
- Multimodal AI Systems with Spiking neural networks

Research & Academic Skills:

- Proven ability in academic writing, producing publication-ready research papers and technical reports for conferences and journals.
- Strong interdisciplinary research experience at the intersection of AI, healthcare, and neuroscience.
- Advanced critical thinking with expertise in hypothesis formulation, experimental design, and data-driven analysis.
- Effective communicator in international research environments, engaging diverse academic and multidisciplinary audiences.

Academic Work Experience:

- Working as a research supervisor and viva panel examiner for undergraduates and master's students. (IIT)
- Conducting Guest Lectures for undergraduate students. (KDU | IIT)
 - Subjects: Data Structures and Algorithms, Machine Learning
- Conducting Workshops for undergraduate students. (KDU | IIT | NSBM)
 - Subjects: Data Structures and Algorithms, Programming Fundamentals

Programming: Java, Python, JavaScript

Algorithms: U-net, CNNs, YOLO, SSDs, Mask R-CNN, K-Means Clustering, Linear Regression

Frameworks: Angular, React, Ionic, QT, TensorFlow, Keras, PyTorch, OpenCV, Spark, MATLAB

Database: MySQL, MongoDB, Elasticsearch, Firebase

DevOps & Infrastructure: Kubernetes, Docker, Terraform, Ansible, AWS, Azure, GCP, CI/CD

Soft: Communication, Time-management, Critical-thinking, problem-solving, Collaboration, teamwork, Leadership, Self-motivation

Industry Work Experience:

Senior Software Engineer, Platform Infrastructure IFS, Oct 2023 - Present

- As a member of the IFS.AI research guild, I contribute and involved with my experience and ideas for AI initiatives.
- Re-designed and maintained the IFS Monitoring product's Bitbucket repository structure, codebase organization, and pipeline architecture for both on-prem and cloud deployments by standardizing CI/CD, test automation, and deployment workflows, and establishing internal technical practice policies.
- Contributed to core architecture MLOps initiatives, integrating monitoring, CI/CD, and safety validation for ML Services.

- Contributed to core architecture initiatives including model inheritance, time zone support, EBR, backward compatibility, and near-zero-downtime upgrades.
- Delivered technical workshops on platform deployment and monitoring for new joiners to R&D.
- Conducted interviews to recruit skilled engineers, and fostered team growth through mentorship and collaborative problem-solving.

Software Engineer, Platform Infrastructure IFS, Mar 2022 - Sep 2023

- Built end-to-end automation capabilities with Site Reliability Engineering teams to reduce SRE on-call alerts by 50% by automating monitoring and alerting for the IFS platform.
- Created comprehensive training materials including instructional videos and documentation for internal teams and external customers, improving product adoption and user satisfaction.
- Managed and maintained on-prem and cloud environments, ensuring 99% uptime for all critical systems.

Associate Software Engineer Zilingo, Jan 2020 - Jan 2021

- Worked on zFactory product that all-encompassing solution for apparel manufacturing an integrated digital ecosystem that enables fast data insights real-time data capturing and visualization.
- Developed and implemented Platform provisioning automation tool application for configuring the entire product client-wise, reducing development time by 80%.

Software Engineer Intern nCinga Innovations, June 2019 – Dec 2019

- Developed and implemented a Java REST API using Elasticsearch and Mongo DB backend micro services to provide real-time data access to a fleet of more than 100 devices.
- Developed and implemented tab-based hybrid mobile applications for capturing data in real-time, which were used by 1000 users to collect over 1 million data points per day.

Extra-Curricular & Memberships:

- Member Of LEO, CTKF & ITKF
- Karate (Japan-Traditional) - National Champion 2014, 2015, 2017

Languages:

- English (Fluent)
- Sinhala (Native)

References:

Dr. Dharshana Kasthurirathna, Assistant Professor at Sri Lanka Institute of Information Technology.
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